LLM 3-oy 5-Days Report.

SFT da datani sonidan ko'ra uni to'g'ri yozilganligi muhim.

Data Source'ni biz Hugging Face dan olsak bo'ladi bu biz nima qilishimizga bog'liq bo'ladi. M: Alpaca, Dolly 15k, Ultra chat kabi datasetlar mavjud, vaziyatdan kelib chiqib post training da foy-sh mumkin.
Data sourcelar maqsadga qarab tanlanadi

Bizni loyihamizda

sft/seeds.txt
sft/sft-500_uzbek-samples.txt
sft/extra_general.txt
sft/external-alpaca_uz.txt
sft/external_oasst_en.txt
sft/seeds.txt
sft/build_sft.py
sft/train.jsonl

fayllarimizdan foy-z

Steps
- Download public data — sft/fetch_external
- Build the dataset — sft/build_sft.py
- Guard against test leaks — ~~sft~~/eval_session

MODEL_SPEC.md fayl Eng Avval yoziladi. Unda
① version
② purpose
③ base model
④ owner
⑤ users
⑥ languages
⑦ Answer style
⑧ Tasks it must do well
⑨ limits
⑩ Not in this version
⑪ Success test
⑫ Change log

eval_set modellarni post training bo'lib tugagach uni turli xil yo'nalishlarda sinaydi.

adapters- yordamida qo'shimcha ulanadigan narsalarni ulab beradi LoRA va QLorani ishlatganimizda joy ochilib qoladi.

SFT da bilishimiz kerak bo'lgan Asosiy narsa data format va data source.

LoRA va QLoRA da Transformer blogi 4 bitga siqiladi

$4 \Rightarrow 2^4 \Rightarrow 16$ 0000 dan 1111

Qloraga cpu b-n hisoblab bo'lmaydi GPU kerak bo'ladi

Demak LoRada bitlar 32 bit QLora da esa 4 bit bo'ladi

Evaluation

eval/eval_set.json — 67 ta test savoli va namunali javoblar, barcha modellar shu bir xil imtixon da tekshiriladi

eval/run_eval.py - bitta modelga 67 ta savolni berib, javoblarini tekshiradi va faylga saqlaydi eval/summarize.py – saqlangan javoblardan barcha modellar uchun solishtirma jadval yasaydi.

eval/results/base.json – base modelning 68 ta javobi, o'qitishdan oldingi boshlang'ich natija (5/68)

eval/results/sft_lora.json – shu kompyuterda CPU da o'qitilgan LoRA modelning javoblari (32/68)

eval/results/sft_qlora.json – Kaggle GPU da o'qitilgan QLoRA modelning javoblari (32/68)

Dataset Manbalari

sft/seeds.txt – qo'lda yozilgan asosiy savol-javoblar, kurs mavzulari, bilmayman rad javoblari, salomlashish va tarjima [illegible]

sft/sft_500_uzbek_samples.txt – [illegible] boshqa model yordamida yaratgan 500 ta namuna; 100 ta tarif, har biri besh xil savol bilan

sft/external_alpaca_uz.txt – Hugging Face dagi o'zbekcha Alpaca datasetidan filtrlab olingan 49 ta namuna

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sft/extra_general.txt – siz qo'shgan umumiy bilim, mantiq, kod va matn yozish namunalari.

sft/external_oasst_en.txt – Open Assistant datasetidan filtrdan o'tgan 240 inglizcha namuna, mavjuddan tashqari bo'lgani uchun datasetga qo'shilma

Dataset yuklaydigan va yasaydigan kodlar.

sft/fetch_external.py – Hugging Face dan ochiq datasetlarni yuklab olib, filtrlaydi va external.txt faylini yoz

sft/build_sft.py – barcha manbalarni yig'adi, arifmetika va tarjima savollarini yasaydi, takror va sifatsiz namunalarni tashlaydi, test savollarini bloklaydi

Dataset Formatlari

Instruction-response (Alpaca uslubi)

Input-output (prompt completion)

Messages (chat)